



PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 05 – Discriminative Classifiers II: Perceptrons and Decision Trees



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One complaint can teach a spam filter

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Scene

A user marks one urgent looking message as spam.

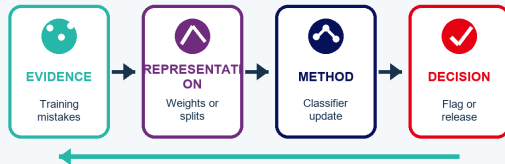
Perceptron response

Move one linear boundary because the current prediction was wrong.

Tree response

Ask which condition should split similar messages next.

THIS LECTURE'S DECISION LOOP



Outcome and error feed the next observation

How do mistakes update a rule & questions partition a space?

Syllabus boundary for Lecture 05

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Syllabus topic	Allocated time	Class emphasis
Perceptron algorithm	1.0 hour	updates, convergence, failure
Decision trees	1.5 hours	partition, split, pruning
Integrated practice	0.5 hour	coding and case reasoning



Deliberate boundary

Feature engineering begins in Lecture 06. Random forests wait until the ensemble lectures.

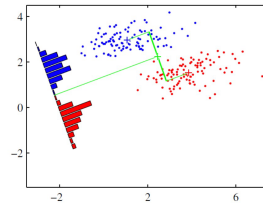
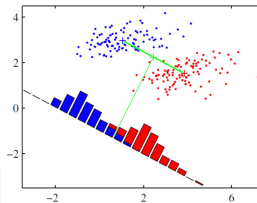
Two boundary languages

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Model	Prediction language	Typical geometry
Perceptron	$\text{sign}(w^T x + b)$	one hyperplane
Decision tree	a path of feature tests	axis aligned regions

Same data

The two models may fit different patterns and fail for different reasons.



A linear score before learning

Score

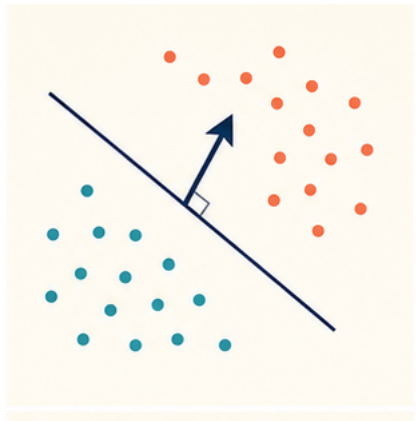
$$a(x) = w^\top x + b$$

Decision

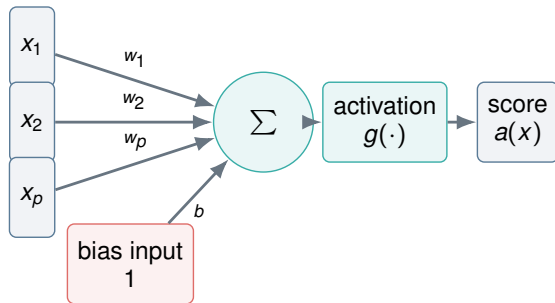
Predict class +1 when $a(x) > 0$, otherwise predict class -1.

Geometric reading

w points across the boundary and b shifts it.



Bias inside an augmented vector



Augmented sample

$$\tilde{X} = [X^\top, \mathbf{1}]^\top$$

Augmented weight

$$\tilde{w} = [w^\top, b]^\top$$

$$a(x) = \tilde{w}^\top \tilde{x}$$

ecome signs 数字工程研究中心

Use $y_i \in \{-1, +1\}$.

$$y_i \tilde{w}^\top \tilde{x}_i > 0$$



A correct prediction makes the signed score positive.

$y_i \tilde{w}^\top \tilde{x}_i > 0$ gives no change.

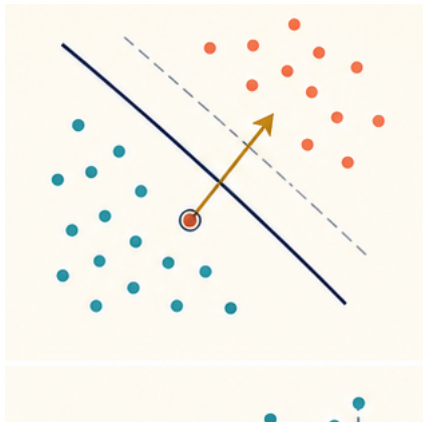
$$y_i \tilde{w}^\top \tilde{x}_i \leq 0$$

$$\tilde{W} \leftarrow \tilde{W} + \eta y_i \tilde{X}_i$$

A positive message lies on the negative side.

Add ηx to w and add η to b .

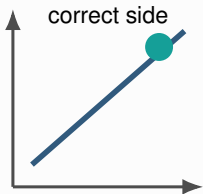
One sample changes the score of every future sample.



Reward and punishment as an early metaphor

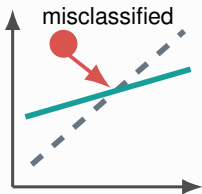
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Correct sample



no update

Mistake



move the boundary

Historical wording

Correct patterns receive no penalty. Misclassified patterns push the weight vector.

Modern reading

The rule optimizes a classification criterion. It does not imitate a biological neuron in detail.

A four point training set

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Sample	$\tilde{x} = (x_1, x_2, 1)$	y
A	(0,0,1)	+1
B	(0,1,1)	+1
C	(1,0,1)	-1
D	(1,1,1)	-1



Initial state

$\tilde{w}^{(0)} = (0, 0, 0)$ and $\eta = 1$.

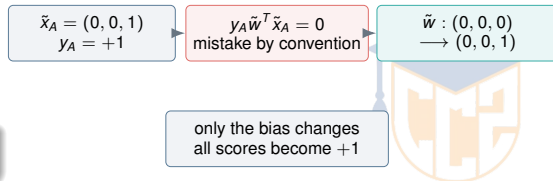
The first mistake creates a bias

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$$y_A \tilde{w}^{(0)\top} \tilde{x}_A = 0$$
$$\tilde{w}^{(1)} = (0, 0, 0) + (0, 0, 1) = (0, 0, 1)$$

Meaning

The classifier now predicts every point as positive.



A negative mistake changes the slope

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$$y_C \tilde{w}^\top \tilde{x}_C = (-1)(1) = -1$$
$$\tilde{w} \leftarrow (0, 0, 1) - (1, 0, 1) = (-1, 0, 0)$$

**Boundary**

$$-x_1 = 0$$

Several epochs can be necessary

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Epoch

One pass through all training samples.

Stopping test

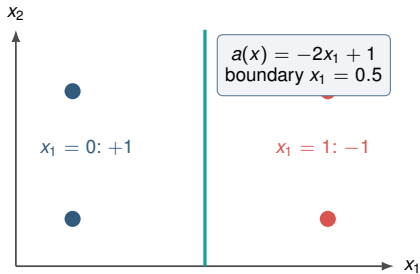
Finish an entire epoch with zero mistakes.

Bad shortcut

Stopping after a lucky local run can leave earlier samples misclassified.

The learned boundary in the classroom example

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The second feature receives zero weight because it is irrelevant in this four-point example.

One valid solution

$$\tilde{w} = (-2, 0, 1)$$

$$a(x) = -2x_1 + 1$$

Check

$x_1 = 0$ gives $+1$ and $x_1 = 1$ gives -1 .

Perceptron learning loop

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```
import numpy as np
X = np.array([[0.,0.,1.],[0.,1.,1.],[1.,0.,1.],[1.,1.,1.]])
y = np.array([1., 1., -1., -1.])
w = np.zeros(3)

for epoch in range(20):
    mistakes = 0
    for xi, yi in zip(X, y):
        if yi * (w @ xi) <= 0:
            w += yi * xi
            mistakes += 1
    print(epoch, mistakes, w)
    if mistakes == 0:
        break
```

Expected pattern

Mistakes decrease to zero for this separable data.

Learning rate changes the path

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Update size

η controls how far one mistake moves the weight vector.

Separable case

A fixed positive learning rate preserves the convergence guarantee.

Finite budget

With early stopping, different learning rates can leave different classifiers.

Sample order changes the solution

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Same constraints

Every training point demands a positive signed score.

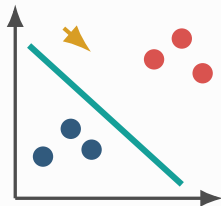
Different path

The next violated constraint depends on sample order.

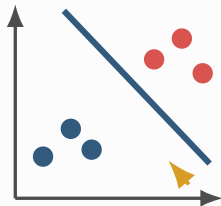
Consequence

Two valid runs can end at different separating hyperplanes.

Order A



Order B



Convergence needs linear separability

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Guarantee

For linearly separable data, repeated mistake updates reach a separating weight vector in finitely many mistakes.

Condition

The guarantee disappears when no separating hyperplane exists.

Margin explains an informal mistake bound

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Data radius

Assume $\|x_i\| \leq R$.

Separating margin

Assume a unit vector u with $y_i u^\top x_i \geq \gamma > 0$.

$$M \leq (R/\gamma)^2$$

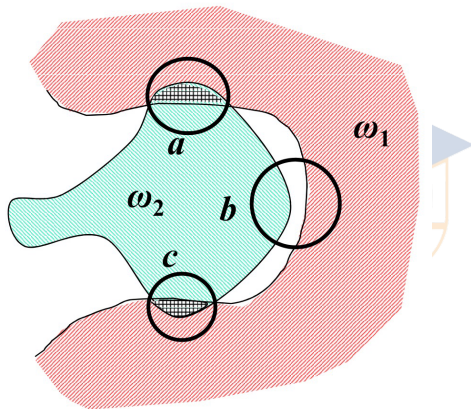
Exclusive OR makes the perceptron cycle

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x_1	x_2	y
0	0	-1
0	1	+1
1	0	+1
1	1	-1

Geometry

No straight line separates the two positive corners from the two negative corners.



Cycling is a diagnostic signal

Observed pattern

Training mistakes repeat without reaching zero.

- 1 Check label errors and duplicate rows with conflicting labels.
- 2 Plot low dimensional projections.
- 3 Try a richer representation or a different model.



Do not claim convergence

A small recent mistake count is not a separating solution.

The pocket rule keeps the best visit

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Idea

During a nonconvergent run, remember the weight vector with the fewest training mistakes.

Benefit

The final epoch no longer automatically decides the returned classifier.

Limit

Training error still needs validation on unseen data.

Averaging reduces path noise

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Averaged perceptron

Return the average of visited weight vectors rather than only the last vector.

Intuition

Short lived oscillations cancel while persistent directions remain.

Still linear

Averaging does not solve a genuinely nonlinear boundary.



Multiclass perceptron uses one score per class

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Scores

$$a_k(x) = w_k^\top x$$

$$\hat{y} = \arg \max_k a_k(x)$$



Interpretation

Each class owns a direction that competes for the sample.

A multiclass mistake updates two rows

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If $\hat{y} \neq y$

$$w_y \leftarrow w_y + \eta X$$

Competing class

$$w_{\hat{y}} \leftarrow w_{\hat{y}} - \eta X$$

Effect

Raise the true class score and lower the mistaken winner.

One versus rest is a different construction

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Construction	Training	Possible issue
Multiclass perceptron	joint competing scores	relative margins
One versus rest	independent binary models	scores may not align



Do not merge the names

Both use linear scores, but their training objectives differ.

Feature scale can dominate updates

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Two features

Age ranges from 18 to 70 while annual spending ranges from 0 to 100000.

Update

The change ηyx inherits the units and scale of every feature.

Consequence

A large numeric range can dominate the path without being more informative.

Bias must follow the same update convention

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Separate form

$$w \leftarrow w + \eta y x \text{ and } b \leftarrow b + \eta y$$

Augmented form

$$\tilde{w} \leftarrow \tilde{w} + \eta y \tilde{x}$$

Common bug

Updating w but forgetting b forces the boundary through the origin.

Prediction at exactly zero needs one rule

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Option A

Treat $a(x) = 0$ as a mistake during training.

Option B

Break the prediction tie with a fixed class convention.

Requirement

Use the same convention in hand calculations, code, and evaluation.

```
def train_perceptron(X, y, epochs=30, eta=1.0):
    w = np.zeros(X.shape[1])
    for epoch in range(epochs):
        mistakes = 0
        for xi, yi in zip(X, y):
            if yi * (w @ xi) <= 0:
                w = w + eta * yi * xi
                mistakes += 1
        if mistakes == 0:
            return w, epoch + 1
    return w, epochs
```

Replace `<=` with `<` and test a zero initialization.

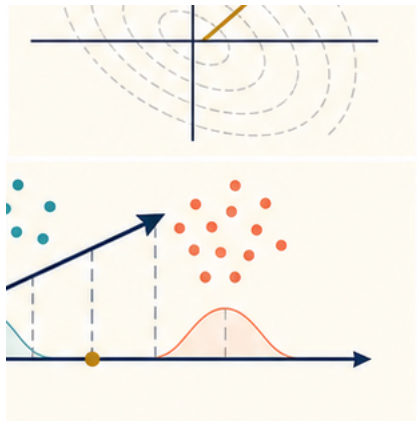
Perceptron lessons that survive the formula

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- Mistakes drive learning.
- Linear separability supports convergence.
- Order and scale affect the path.
- A zero training error does not guarantee future performance.

Model boundary

A linear learner needs a richer representation when the pattern is nonlinear.



A hospital triage tree

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Decision-tree case: from split to action

First question

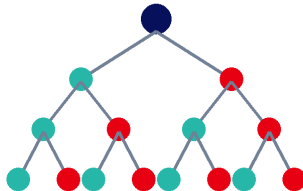
Is oxygen saturation below 92 percent?

Second question

If not, is breathing rate above the age adjusted threshold?

Leaf action

Urgent review, routine observation, or discharge advice.



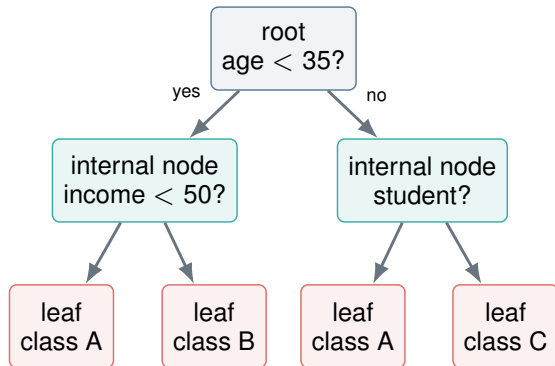
Trace one sample

Report the leaf
distribution

Test nearby
counterfactuals

Root, internal nodes, branches, and leaves

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Root

The first test sees every training sample.

Internal node

A feature condition routes a subset.

Leaf

A prediction summarizes the samples that arrive there.

A branch records a test outcome; a leaf returns the prediction.



Low oxygen and high breathing rate lead to urgent review.

Conditions on different branches do not all apply to one sample.

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A classification leaf stores a class distribution

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$$\hat{p}_{mk} = \frac{n_{mk}}{n_m}$$
$$\hat{y}_m = \arg \max_k \hat{p}_{mk}$$



Small leaf

A pure leaf with two samples gives a fragile probability estimate.

$$\hat{c}_m = \frac{1}{n_m} \sum_{i: x_i \in R_m} y_i$$

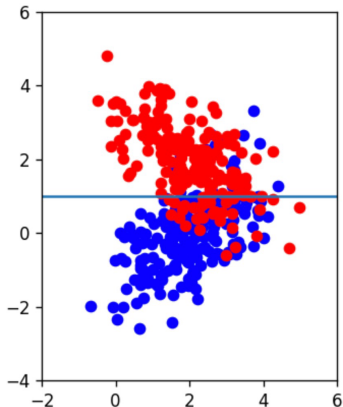
$$\hat{f}(x) = \sum_m \hat{c}_m \mathbb{I}(x \in R_m)$$

The prediction changes in steps when a sample crosses a split.

A split reads like a simple threshold rule.

A diagonal boundary may require many small rectangles.

Construct a rotated feature or use a model with oblique boundaries.



Payment method has card, cash, transfer, and voucher values.

Valid split

Send a subset of categories left and the remaining categories right.

Invalid shortcut

Integer codes do not create a meaningful order for nominal categories.

Impute before fitting the tree.

Use a model that learns a default branch for missing values.

Missingness may itself carry information and may change after deployment.

The same binary questions can separate several classes step by step.

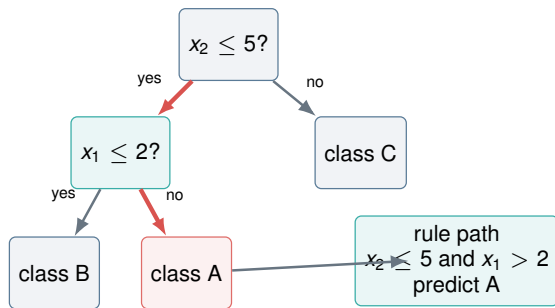
Each leaf may contain counts for every class.

Use case

A fault diagnosis tree can end in normal, sensor fault, motor fault, or overload.

A borrowed tree example becomes a rule list

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Read the diagram

Start at the top test, follow one branch result, and stop at a leaf label.

Translation

If $x_2 \leq 5$ and then $x_1 > 2$, predict the leaf class shown on that path.

Overclaim

The path does not prove that changing debt ratio will cause the outcome to change in reality.

Tree rules translate directly into code

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```
def manual_tree(oxygen, breathing_rate):  
    if oxygen < 92:  
        return "urgent review"  
    if breathing_rate > 24:  
        return "observe"  
    return "routine advice"
```

Trace

oxygen=95 and breathing_rate=27 reaches the second return.

Depth	Typical behavior	Risk
1	one stump	underfitting
3 to 5	short interactions	often readable
very deep	small specialized leaves	high variance

Validation data must judge the tradeoff.

Deep trees have low bias and high variance

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Low bias

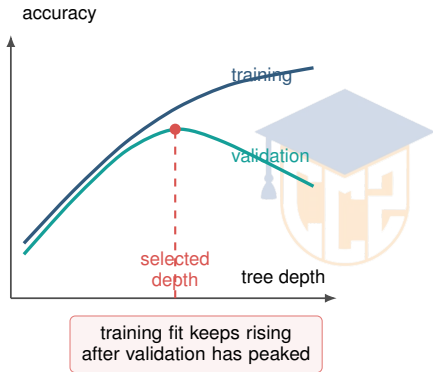
A sufficiently deep tree can fit intricate training patterns.

High variance

Small data changes can alter early splits and the entire subtree below them.

Symptom

Training accuracy stays high while validation accuracy falls.



The best root split is oxygen < 92.

After one borderline case changes, breathing rate becomes the best root split.

A path can look precise even when the tree structure is unstable.

Stopping rules limit fragmentation

- Maximum depth limits path length.
- Minimum samples for a split limits tiny parent nodes.
- Minimum samples per leaf limits fragile terminal groups.
- Minimum impurity decrease rejects weak splits.

Tune together

Different rules can produce similar sized trees for different reasons.

Minimum leaf size protects probability estimates

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Leaf counts	Estimated positive rate	Reliability
2 of 2	1.00	very uncertain
40 of 50	0.80	more stable
400 of 500	0.80	still population dependent



Same purity, different evidence

Count the observations behind every leaf probability.

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A shallow tree in Python

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```
from sklearn.tree import DecisionTreeClassifier

tree = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    min_samples_leaf=20,
    random_state=7,
)
tree.fit(X_train, y_train)
print(tree.get_depth(), tree.get_n_leaves())
print(tree.score(X_train, y_train))
print(tree.score(X_valid, y_valid))
```

Read together

Depth, leaf count, training score, and validation score describe different parts of the fit.

Text rules expose the fitted structure

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```
from sklearn.tree import export_text

rules = export_text(tree, feature_names=list(feature_names))
print(rules)
```

Inspection question

Do the root feature and thresholds make sense in the measurement units?

Tree tracing checkpoint

Applicant

income=8, debt ratio=0.62, prior defaults=0

- 1 Follow the root test.
- 2 Write every condition on the selected path.
- 3 Report the leaf sample count and class ratio.
- 4 Name one nearby change that reaches a different leaf.



Complete answer

A class label without a path and leaf evidence is incomplete.

A useful split makes children more homogeneous

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Parent node

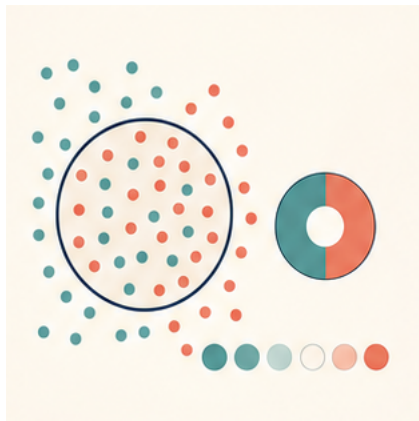
Several classes are mixed together.

Candidate split

A feature threshold sends samples into two children.

Goal

The weighted child impurity should be lower than the parent impurity.



Entropy measures label uncertainty

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$$H(R) = - \sum_{k=1}^K p_k \log_2 p_k$$

Pure node

$H = 0$ when one class has probability 1.

Balanced binary node

$H = 1$ bit when $p_1 = p_2 = 0.5$.

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$$G(R) = 1 - \sum_{k=1}^K p_k^2$$

$$G = 2p(1 - p).$$

A pure node gives 0. A balanced binary node gives 0.5.

Useful for describing leaf error.

It may treat distinct probability mixtures as equally impure.

Positive share	Entropy	Gini	Error
0.50	1.000	0.500	0.500
0.80	0.722	0.320	0.200
1.00	0.000	0.000	0.000

All three rank these nodes from most mixed to pure.

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The split removes much label uncertainty.

Child sizes belong in the formula.

Gini decrease follows the same weighting pattern

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$$\Delta G = G(R) - \frac{n_L}{n} G(R_L) - \frac{n_R}{n} G(R_R)$$



Tree choice

Select the feature and threshold with the largest decrease among current candidates.

A hand calculation for one split

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Node	Positive	Negative	Gini
Parent	4	4	0.500
Left	3	1	0.375
Right	1	3	0.375

$$\Delta G = 0.5 - \frac{4}{8}(0.375) - \frac{4}{8}(0.375) = 0.125$$



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Child	Size	Positive share
Left	1	1.00
Right	99	0.49

The pure child looks impressive but represents only one observation.

Use the weighted impurity decrease and minimum leaf constraints.

Consider midpoints between adjacent distinct sorted values.

For 2, 3, 7, and 10, candidates include 2.5, 5, and 8.5.

Do not split identical values into different branches.

Recount every class on both sides for every threshold.

Move one sorted sample at a time and update left and right class counts.

Result

Every threshold can be scored in one pass after sorting.

A feature with K categories has many possible binary partitions.

Exhaustive subset search becomes expensive when cardinality is high.

Group rare categories, use ordered statistics carefully, or choose a model with native categorical handling.

Classification and regression trees use binary splits

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Classification

A classification tree commonly minimizes Gini impurity.

Regression

A regression tree commonly minimizes squared error within children.

Structure

Every internal node has two outgoing branches.

Regression splits reduce within node squared error

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$$E(R) = \sum_{i: x_i \in R} (y_i - \bar{y}_R)^2$$

$$\Delta E = E(R) - E(R_L) - E(R_R)$$



Leaf prediction

Each child predicts its own mean.



Tree structures, features, thresholds, and stopping points interact.

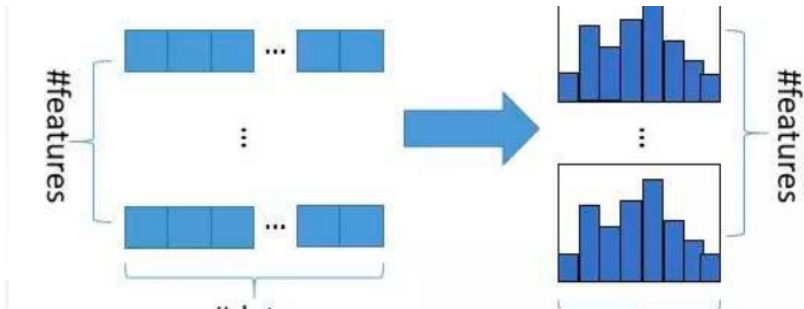
Grow greedily, constrain complexity, and validate or prune.

1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

The fitted tree is a useful heuristic solution, not proof of a globally optimal structure.

Histogram splits trade precision for speed

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Binning. Map continuous values into a limited number of bins.

Search. Accumulate class or gradient statistics per bin, then test bin boundaries.

Benefit. Large datasets need fewer candidate thresholds.

A split search in plain Python

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```
def gini(y):
    if len(y) == 0:
        return 0.0
    p = np.bincount(y, minlength=2) / len(y)
    return 1.0 - np.sum(p ** 2)

def split_gain(x, y, threshold):
    left = x <= threshold
    n = len(y)
    child = left.sum()/n * gini(y[left])
    child += (~left).sum()/n * gini(y[~left])
    return gini(y) - child
```

Unit test

A perfect separation should give a larger gain than a random threshold.

Exclusive OR is easy for a small tree

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First split

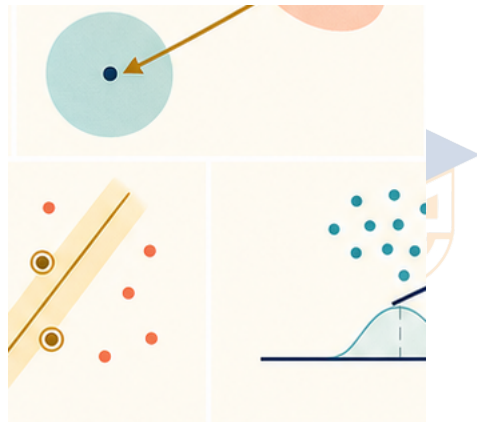
Test $x_1 \leq 0.5$.

Second level

Within each side, test $x_2 \leq 0.5$.

Result

Four leaves represent the four corners exactly.



A fully grown tree can memorize noise

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Training data

A mislabeled customer becomes a single pure leaf.

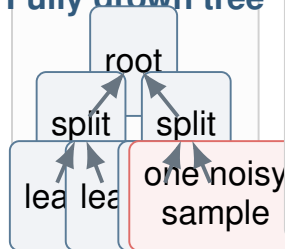
Future data

The special rule has no stable support and fails on similar customers.

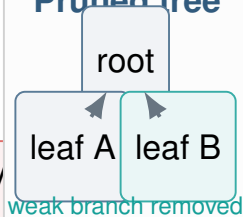
Response

Remove weak branches or prevent them from growing.

Fully grown tree



Pruned tree



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Depth, leaf size, split size, and minimum impurity decrease.

Training remains faster and the tree stays compact.

Risk

An early weak split may enable valuable later structure, so stopping can be premature.

Grow a relatively rich tree.

Evaluate nested subtrees and remove branches whose benefit does not justify their complexity.

Later evidence can judge whether an early split became useful.

Cost complexity pruning balances fit and size

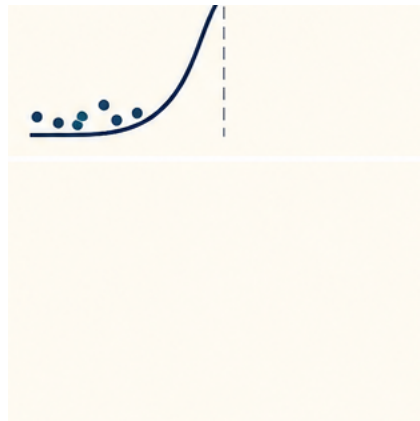
$$R_{\alpha}(T) = R(T) + \alpha |T_{\text{leaves}}|$$

Fit term

$R(T)$ measures training error or impurity.

Complexity term

α charges every leaf.



Keep more branches.

Collapse the weakest link and obtain a smaller subtree.

Candidate trees are nested rather than unrelated random structures.

Pre-pruning

Post-pruning

Validation selects complexity

Validation selects the subtree

- 1 Fit candidate pruning paths on training folds.
- 2 Evaluate each candidate on held out folds.
- 3 Choose the complexity rule.
- 4 Refit on the available training data.
- 5 Evaluate once on the untouched test set.

Leakage

Pruning on the test set turns the test set into training data.

Training and validation curves tell different stories

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Tree size	Training error	Validation error
small	high	high
moderate	lower	lowest
very large	near zero	higher



Selection

The validation minimum identifies useful complexity, not the training minimum.

Leaf probabilities often need calibration

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Raw estimate

Class frequency within a leaf.

Problem

Deep leaves produce extreme 0 or 1 estimates from few samples.

Evaluation

Use calibration curves and proper scoring rules on validation data.

Laplace smoothing tempers tiny leaves

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$$\tilde{p}_k = \frac{n_k + \alpha}{n + K\alpha}$$

Binary leaf

For 2 positives out of 2 and $\alpha = 1$, the estimate becomes 3/4 rather than 1.

Limit

Smoothing helps small counts but does not fix dataset shift.

Impurity importance can favor flexible features

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Mechanism

Add impurity decreases from every split that uses a feature.

Bias

Continuous or high cardinality features offer more candidate splits and may appear important by chance.

Check

Compare with validation permutation importance and domain knowledge.

Tree explanations need nearby counterexamples

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Path explanation

List the conditions that led to the leaf.

Counterexample

Change one feasible feature and trace whether the route changes.

Feasibility

A mathematically small change may be impossible in the real system.

Bootstrap trees reveal structural stability

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Procedure

Refit the same tree specification on several bootstrap samples.

Inspect

Track root features, common paths, and validation performance.

Finding

Similar scores with different roots signal predictive usefulness but weak structural certainty.

CASE EVIDENCE TO ACTION

**OBSERVE**

Training mistakes

**USE**

Pruning & honest tree evaluation

FAILURE TO TEST**RESOLVE**

Why can a smaller tree generalize better?

Pruning on the final test set.

Perceptron and tree assumptions

Question	Perceptron	Decision tree
Boundary	one hyperplane	rectangular regions
Learning signal	misclassified samples	impurity decrease
Main risk	nonseparability	high variance
Typical control	representation and stopping	depth and pruning

Choice

Match the model language to the pattern and the cost of mistakes.

One dataset can support two useful views

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Perceptron view

A global weighted score summarizes risk.

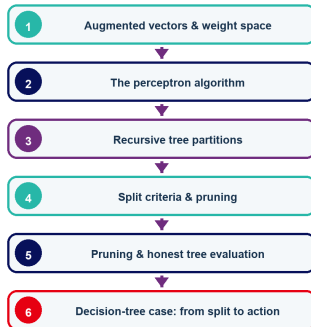
Tree view

A short sequence of conditions separates operational groups.

Comparison

Use the same train, validation, and test partitions for a fair contest.

SIX MODULES, ONE ARGUMENT



Coding comparison on the same split

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```
from sklearn.linear_model import Perceptron
from sklearn.tree import DecisionTreeClassifier

models = {
    "perceptron": Perceptron(max_iter=1000, random_state=7),
    "tree": DecisionTreeClassifier(max_depth=3, min_samples_leaf=10,
                                   random_state=7),
}
for name, model in models.items():
    model.fit(X_train, y_train)
    print(name, model.score(X_train, y_train),
          model.score(X_valid, y_valid))
```

Next step

Inspect errors and structure after the score comparison.

Integrated case for an inspection gate

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Data

Temperature, vibration, operating mode, and 300 labeled inspections.

- 1 Build a scaled perceptron baseline.
- 2 Build a shallow pruned tree.
- 3 Compare costly missed faults.
- 4 Inspect stability after resampling.



Deployment

Record how missing sensors route through the system.

Failure diagnosis by symptom

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Symptom	Likely cause	First check
Perceptron cycles	not separable or bad labels	plot and audit labels
Tree trains perfectly	excess complexity	validation gap and leaf sizes
Tree root keeps changing	correlated features or small data	bootstrap stability
High score disappears	leakage or shift	pipeline and time split

Perceptron and tree self-check

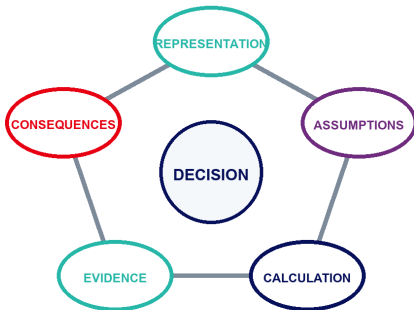
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- 1 Explain one perceptron update in words.
- 2 State the condition behind perceptron convergence.
- 3 Compute a weighted Gini decrease.
- 4 Translate one root to leaf path into a rule.
- 5 Explain why pruning must not use the test set.

Strong answer

Use a concrete sample, one equation, and one failure condition.

A CHECKABLE DECISION LOOP



How do mistakes update a rule & questions partition a space?

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We learned how two classifiers use a feature vector.

We will decide how raw values become valid, useful, and leakage free features.

Bridge

The same raw timestamp can become an hour, a weekday, a cyclic pair, or a source of leakage.